Calculation of intrathecal κ-FLC synthesis

Different approaches that consider serum κ-FLC concentrations and/ or blood-CSF-barrier function have been used to calculate an intrathecal κ-FLC synthesis. Blood-CSF-barrier function was assessed by the CSF/ serum albumin quotient (Q_{alb}).

(1) κ-FLC quotient

$$Q_{\kappa-FLC} = \frac{CSF\ \kappa\text{-}FLC}{Serum\ \kappa\text{-}FLC} \qquad \text{Eq. (1)}$$

The presence of an intrathecal κ-FLC synthesis is determined by referring each patient's $Q_{\kappa\text{-}FLC}$ to a pre-defined cut-off. The extent of the κ-FLC synthesis is reflected by the exact $Q_{\kappa\text{-}FLC}$ value.

(2) κ-FLC index

$$\kappa - FLC\ index\ = \frac{\kappa-FLC_{CSF}\ /\ \kappa-FLC_{Serum}}{Q_{alb}} \qquad \text{Eq.(2)}$$

The presence of an intrathecal κ-FLC synthesis is determined by referring each patient's *κ-FLC index* to a pre-defined cut-off. The extent of the κ-FLC synthesis is reflected by the exact index value.

(3) Intrathecal κ-FLC fraction

The presence and extent of an intrathecal κ-FLC synthesis is calculated by a non-linear function relating each $Q_{\kappa\text{-}FLC}$ to its corresponding Q_{alb}-dependent upper normal limit ($Q_{lim\ \kappa\text{-}FLC}$). $Q_{lim\ \kappa\text{-}FLC}$ has been previously defined by different formulae:

Presslauer 2014	$Q_{lim\ \kappa-FLC} = 0.9357 \cdot Qalb^{0.6687}$	Eq. (3)
Hegen 2019	$Q_{lim\ \kappa-FLC} = 3.1276 \cdot Qalb^{0.8001}$	Eq. (4)
Senel 2019	$Q_{lim\ \kappa-FLC} = 9.50 + 2.08 \cdot Q_{alb}$	Eq. (5)
Reiber 2019	$Q_{lim\ \kappa-FLC} = 3.27 \cdot \left(Q_{alb}{}^2 + 33\right)^{0.5} - 8.2 \cdot 10^{-3}$	Eq. (6)

Locally produced κ-FLC concentration in CSF is calculated as the difference of $Q_{\kappa\text{-}FLC}$ to $Q_{lim\ \kappa\text{-}FLC}$ and corrected for absolute κ-FLC serum concentration:

$$\kappa - FLC_{Loc} = (Q_{\kappa-FLC} - Q_{\lim \kappa-FLC}) \cdot \kappa - FLC_{Serum} \quad \text{Eq. (7)}$$

Finally, the relative intrathecal κ-FLC fraction ($IF_{\kappa\text{-}FLC}$) is displayed as percentage according to following formula:

$$IF_{\kappa-FLC} = \frac{\kappa-FLC_{Loc}}{\kappa-FLC_{CSF}} \cdot 100 \quad \text{Eq. (8)}$$